Baekcheon Seong

Senior Optical Engineer at Koh Young Technology

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Professional Summary

Optical Engineer with a Ph.D. and 3+ years of industry R&D at Koh Young Technology. Core expertise: bridging advanced optical theory and robust industrial systems through optical system design and computational algorithms.

At Koh Young I develop computational inspection systems: On-the-fly Multi-View Photometric Stereo (MVPS) and depth-of-field (DoF) extension for **AOI** and **wire-bonding** inspection (**lead**); line scan fluorescence imaging for **PID** with 2 µm pitch **RDL** on 300 mm wafers (**lead**); optimized digital holography algorithms for BGA micro-bump metrology. I drive **requirements**, **calibration**, and field support from concept to mass production; strengths in **rapid prototyping** and **system-level troubleshooting**.

Ph.D. from the **Computational Imaging & Instrumentation Laboratory** (Yonsei); academic focus on physics-informed deep learning and end-to-end optimization (e.g. phase-contrast imaging, depth-of-field extension via pupil engineering).

Professional Experience

Koh Young Technology, Yongin, South Korea . . . *2023–Present*

Senior Engineer, Optical Group

- *Global leader in 3D measurement and inspection technology (solder paste inspection / automated optical inspection, SPI/AOI).*
- **On-the-fly Multi-View Photometric Stereo for AOI** . . . 2024–Present
  - **Independently proposed and validated an inspection concept**: verified algorithms with synthetic data (Blender-Python), then demonstrated feasibility with a hands-on prototype; secured its adoption as a cross-functional project. **Led concept validation, prototype development, and technical hand-off** to the product team.
  - **Solely constructed a fully functional prototype** using off-the-shelf optics and developed integrated control software (Python, OpenCV) to demonstrate acquisition and reconstruction (motion control, acquisition).
  - For wire-bonding and Automated Optical Inspection (AOI), **developed and led implementation of core 3D algorithms** (Python) utilizing on-the-fly multi-view stereo and Photometric Gradient Fusion for micron-scale shape measurement and for specular surface reconstruction.
  - **Provided** reconstruction **pipelines and datasets** to the Vision group for C++ product port; **defined hardware specifications** (camera, lens, motion) from Blender simulations and prototype experiments; enabled parallel development and shortened development cycle.
  - **Developed production calibration** logic and **supported field engineers** with Python tools and documentation for **production handoff**.
- **Extended Depth-of-Field for Wire Bonding Inspection** . . . 2023–Present
  - Led project applying **Pupil Engineering for wire bonding inspection** in micron-scale Fringe Projection Profilometry; Optical character recognition (OCR) of AOI device in scope. **Solution deployed** as optional module on **production line**.
  - **Implemented** simulation-based **phase plate and post-processing joint optimization** (ZEMAX-MATLAB) using particle swarm optimization to design optimal pupil filters; **achieved real-time processing** by implementing jointly-optimized post-processing in camera FPGA.
  - **Devised a phase plate manufacturing method** to mitigate artifacts under specular reflection, solving a key limitation of pupil engineering in industrial inspection.
- **Line Scan Fluorescence Imaging for PID with 2 µm Pitch RDL** . . . 2023–2024
  - Observed distinct **fluorescence penetration characteristics in Photo-imageable dielectric (PID)** with 2 µm pitch RDL multi-layer stacks on 300 mm wafers, dependent on excitation wavelength.
  - **Designed and prototyped** a multi-wavelength **line-scan fluorescence imaging system** using off-the-shelf optics to resolve signal ambiguities caused by PID thickness variations, utilizing wavelength-dependent penetration to **isolate layer structure**, and demonstrated acquisition and reconstruction with OpenCV-Python.
- **High-Throughput Vertical Scanning Interferometry for Micro-bump Inspection** . . . 2023–2024
  - Conceptualized and **prototyped a high-speed vertical scanning interferometry (VSI)**; optimized light source to **minimize scanning steps** and constructed system from off-the-shelf components (Mirau objectives).
  - Achieved **∼10× scan speed improvement** over Nyquist sampling by applying sampling-theory-based envelope estimation; validated through hands-on experiments.
- **High-Throughput Digital Holography for Micro-bump Inspection** . . . 2023–2024
  - **Optimized** multi-wavelength **digital holography algorithms** (Phase Unwrapping, Noise Reduction); identified root causes (ghosting, wavelength drift, wavefront curvature mismatch) of optical-component issues, validated with VirtualLab, and **addressed and resolved** them by **redefining component specifications**.
  - Validated algorithm performance on micro-bump datasets; provided verified Python algorithms to Vision group for C++ product port.

Yonsei University, Seoul, South Korea . . . *2018–2023*
Graduate Researcher, Computational Imaging & Instrumentation Lab

- **Physics-Informed Optical Design: Pioneered an E2E** optics and post-processing co-design framework, treating the physical optical element as a trainable layer; demonstrated in fluorescence and AOI (*Light Sci. Appl.*, 2023; US Pat. 12,541,819). **Developed an Untrained Deep Learning approach** utilizing physical image formation models; demonstrated in LED-based phase-contrast imaging (*Opt. Lett.*, 2023). **Applied Topology Optimization** to design optics in Fourier domain for focus elongation (*Struct. Multidiscip. Optim.*, 2020). **Tools:** Python, LabVIEW.
- **Advanced Optical System: Built** LSFM with spatial light modulator and electrically tunable lens, demonstrated on biomedical specimens (*Biomed. Opt. Express*, 2020). **Built** Fourier-domain OCT: SD-OCT for polymer film inspection (US Patent 11,846,587), SS-OCT for biomedical applications (*Theranostics*, 2025). **Tools:** Python, LabVIEW. All systems developed in-house with off-the-shelf optics on optical tables.
- **Optical Component Design: Designed Diffractive Optical Elements** (Binary Phase Filters, Fresnel Zone Plates) for diverse imaging applications (*IEEE Trans. Med. Imaging*, 2025; *Opt. Express*, 2020). **Tools:** Zemax–MATLAB API.

Skills

Core competencies: Computational imaging, system integration, rapid prototyping, troubleshooting, technical ownership.
Simulation & design: Zemax, VirtualLab, Blender, CAD (UGNX, Creo, Fusion 360).
Algorithms & software: Python, MATLAB, LabVIEW, git/bitbucket, Linux/Unix.
Optical systems: Industry: multi-view photometric stereo (MVPS), pupil engineering, fluorescence imaging, vertical scanning interferometry (VSI), digital holography; Academic: optical coherence tomography (OCT), light sheet fluorescence microscopy (LSFM), differential phase contrast (DPC), Fourier ptychographic microscopy (FPM).
Domain: SMT and back-end semiconductor inspection (e.g. BGA, wire bonding, 2 µm pitch RDL with PID).
Languages: Korean (Native), English (Business-fluent)

Education

Yonsei University, Seoul, South Korea
Ph.D. in Mechanical Engineering
(Computational Imaging & Instrumentation Lab), 2018–2023

Hongik University, Seoul, South Korea
B.S. in Mechanical Engineering, 2014–2018

Selected Patents

- **Apparatus for Three-Dimensional Shape Measurement**. US Patent Application 20250207912 A1 (2025).
- **Method for Designing Binary Phase Filter**. US Patent 12,541,819 B2 (2023).
- **Tomography Imaging System for Transparent Material Composite Thin Film**. US Patent 11,846,587 B2 (2023).

Selected Peer-Reviewed Publications

- J. Yun, J. Lee, S. Kim, **B. Seong**, Y. Kim, C. H. Yoon, C. Joo, and K. H. Kim. “Oblique Light Sheet Fluorescence Microscopy with Surface Tracking for High-speed Large-area Imaging of Conjunctival Goblet Cells.” *IEEE Trans. Med. Imaging*, (2025).
- S. Han, I. Kim, **B. Seong**, W. Kim, C. Joo, and S. Park. “Dual-parameter tomographic imaging of attenuation and backscattering coefficients for quantitative evaluation of immune cell-mediated cytotoxicity in tumor spheroids.” *Theranostics*, 15(18), 9399 (2025).
- **B. Seong**, W. Kim, Y. Kim, J. Lee, J. Yoo, and C. Joo. “E2E-BPF microscope: extended depth-of-field microscopy using learning-based implementation of binary phase filter and image deconvolution.” *Light Sci. Appl.*, 12, 269 (2023).
- **B. Seong**, I. Kim, T. Moon, M. Ranathunga, D. Kim, and C. Joo. “Untrained deep learning-based differential phase-contrast microscopy.” *Opt. Lett.*, 48(13), 3607–3610 (2023).
- S. Song, J. Kim, T. Moon, **B. Seong**, W. Kim, C. Yoo, J. Choi, and C. Joo. “Polarization-sensitive intensity diffraction tomography.” *Light Sci. Appl.*, 12, 124 (2023).
- R. Han, J. Lee, **B. Seong**, R. Shin, D. Kim, C. Park, J. Lim, C. Joo, and S. Kang. “Direct replication of a glass micro Fresnel zone plate array by laser irradiation using an infrared transmissive mold.” *Opt. Express*, 28(12), 17468–17480 (2020).
- S. W. K. Roper, S. Ryu, **B. Seong**, C. Joo, and I. Y. Kim. “A topology optimization implementation for depth-of-focus extension of binary phase filters.” *Struct. Multidiscip. Optim.*, 62, 2731–2748 (2020).
- S. Ryu, **B. Seong**, C. Lee, M. Y. Ahn, W. T. Kim, K. M. Choe, and C. Joo. “Light sheet fluorescence microscopy using axi-symmetric binary phase filters.” *Biomed. Opt. Express*, 11(7), 3936–3951 (2020).

Selected Conference Presentations

- C. Joo, S. Song, T. Moon, W. Kim, and **B. Seong**. “Computational polarization microscopy for multidimensional and multicontrast imaging.” *SPIE Computational Optical Imaging and Artificial Intelligence in Biomedical Sciences II*, (2025).
- C. Joo, **B. Seong**, Y. Kim, and J. Cho. “Depth-enhanced computational microscopy via co-learned binary phase filter and image deconvolution.” *SPIE Three-Dimensional Imaging, Visualization, and Display 2025*, (2025).
- **B. Seong**, W. Kim, J. Cho, and C. Joo. “Extended depth-of-field microscopy using jointly-learned binary phase filter and image deconvolution.” *SPIE Imaging, Manipulation, and Analysis of Biomolecules, Cells, and Tissues XXI*, (2023).